

Opacity as Commitment: Fragile Equilibria under Delegated Trading*

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June 18, 2025

Abstract

We study a model à la ? in which financial traders care not only about profits but also their reputation, i.e., a recruiter’s perception of their skills. The model presents two self-fulfilling equilibria. One resembles Kyle’s original equilibrium. In the other, a “reputation-driven” equilibrium, traders trade more aggressively to appear skilled, leading to poorer performance and higher price volatility. We show that (1) labor mobility in finance generates traders’ reputation concerns, triggering market fragility and increasing wage inequality, (2) traders with different skill levels adopt different trading styles, and (3) skills in active management may not translate into performance.

JEL classification: G11, G12, G14, G23

Key words: informed trading, market reputation, trader skill, financial fragility, poaching, asset prices

*For helpful comments and discussions, we thank Zhi Da, Paul Ehling, Albert Kyle, Yasuto Monden, Don Noh, Yoshihiro Ohashi, Maureen O’Hara, Wataru Ohta, Norman Schü, and seminar participants at Economic Theory and Policy Workshop (Tokyo), Hitotsubashi University, Nippon Finance Association, Osaka University, SFI Research Days, Summer Workshop on Economic Theory (SWET), Tokyo Metropolitan University, and University of Lausanne. This work is supported by JSPS KAKENHI Grant Number JP17H07084. The views and conclusions expressed in this paper are strictly those of the authors. They do not necessarily reflect those of Peritus Investment Consultancy. Peritus Investment Consultancy takes no responsibility for any errors or omissions in the information contained in this paper. All errors and omissions are our own.

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Conflict-of-interest disclosure statement

Jun Aoyagi: I have nothing to disclose.

Yuki Sato: This work was supported by KAKENHI (Grants-in-Aid for Scientific Research) of JSPS (Japan Society for the Promotion of Science), FY 2017-2019. The Grant Number is JP17H07084, and the total amount is JPY 2,100,000 (about USD 14,000).

Raluca Toma: I have nothing to disclose.

1 Introduction

Financial firms compete for skilled employees. As the boundaries between investment banking, commercial banking, and asset management blur, this competition intensifies. Following the 2007-2009 crisis, pressures to reduce compensation and the resulting challenges in retaining talent have brought poaching practices into focus.¹ Poaching, or more generally labor mobility, raises concerns among financial specialists about their reputation within the industry. How do these concerns influence their trading strategies and skill acquisition? How do they interact with asset prices and market liquidity?

To address these questions, we study a model inspired by [?](#) , where traders, privately informed about their skill, trade risky assets anonymously with market makers. We define skill as the trader-specific ability to uncover more information about asset payoffs than the market. The more skilled a trader is, the larger the potential mispricing she can identify, and thus, the greater the trading profits she can potentially earn. Each trader selects an asset based on her skill and strategically submits a market order to the market maker for that asset, who sets the price after observing the aggregate order flow. A recruiter attempts to infer each trader’s skill from the observed price and payoff of the asset chosen by the trader. The trader values not only trading profits but also her reputation, i.e., the recruiter’s estimate of her skill. In a simple one-period model presented in Section [??](#), each trader derives an exogenous benefit from her reputation, which we model as an increasing quadratic function of reputation. In Section [??](#), we extend the model to a two-period setting, providing a microfoundation for the reputation benefit. In this extension, each trader randomly encounters a recruiter who compensates her for her skill based on his estimate. In equilibrium, the quadratic form assumed in Section [??](#) arises

¹According to *The Wall Street Journal*, poaching is a common practice in the sophisticated capital markets industry, occurring in both bull and bear markets (“[Greed is Good](#),” February 2009), and employees at all levels are being poached (“[Employee Poaching Returns to Wall Street](#),” May 2010). *The New York Times* (“[A Mad Scramble for Young Bankers](#),” July 2014) and *Bloomberg* (“[Saturdays Off Lose to Cash as Buyout Firms Poach Junior Bankers](#),” March 2014) highlight the buy-side’s (private equity, hedge funds) recruitment of young talent from investment banking amid a changing financial industry.

endogenously.

The model yields two stable, self-fulfilling equilibria, one of which is similar to that of ?, where reputation is irrelevant. In the other, the “reputation-driven” equilibrium, traders trade more aggressively to appear skilled. This strategy increases trading volume, price volatility, and market liquidity but undermines trading profits by revealing too much private information to the market maker. Intuitively, each trader can boost her reputation by placing a large order, either buy or sell, because such an action makes her appear like a skilled trader who has identified an asset with significant mispricing. In equilibrium, the recruiter accurately anticipates the trader’s strategy, so the recruiter’s learning is not manipulated; nevertheless, the trader may trade aggressively depending on the market’s belief. To illustrate the mechanism, suppose the recruiter believes that the trader trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase profits, the recruiter (incorrectly) infers that she is less skilled. To avoid the reduction in her reputation benefit, she conforms to the recruiter’s belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the reputation-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.² Key to our multiple equilibria and fragility results are the trader’s conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to the recruiter to obtain the reputation benefit. If the trader’s concern for reputation is very weak (very strong), the first (second) motive dominates, and the Kyle-like (reputation-driven) equilibrium holds uniquely. However, if the concern is moderately strong, either equilibrium may arise depending on the market’s self-fulfilling beliefs. Therefore, the model predicts that fragility may occur

²As in ?, we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

when poaching is (moderately) active and there is high demand for financial skill, as these situations increase the likelihood of future recruitment and lead to heightened reputation concerns.

Our results shed light on how traders’ underlying skills relate to their trading styles and contribute to financial fragility. The model predicts that high-skilled traders adopt low-intensity, profit-driven strategies, while those with lower skills engage in aggressive, risk-seeking behavior to gain reputation. Traders with intermediate skill levels, facing multiple equilibria, may switch between profit-seeking and reputation-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders’ wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader’s skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by ? and ?, who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., ?; ?).

In Section ??, we study traders’ endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a “middling” level of skill, which is too high or too low compared to the

level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. ? is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In ?, strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders' forward-looking concerns about future recruitment opportunities. In ?, learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as ? and ?, in which strategic complementarities play a key role. In ? frenzies originate from complementarities in information markets, while in ? strategic complementarities, triggered by a feedback effect from financial markets to a firm's real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. ?, ?, and ? de-

rive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. ? show that career concerns amplify price volatility, similar to our findings. In ?, career concerns lead to churning (trading without private information), and in ?, they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section ?? presents a baseline model with exogenous reputation concerns, and Section ?? studies the equilibrium. Section ?? endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section ?? analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la ? with delegated investments. The economy consists of a risk-neutral *fund manager*, a competitive risk-neutral *market maker*, *noise traders*, and $n \geq 2$ risk-neutral *investors*. The manager leverages her skill (defined later) to discover a single risky asset with potential mispricing, raises funds from investors, and trades the asset with the market maker who set the price p . The remaining funds are allocated to risk-free savings with a zero interest rate.³

Skill. The manager can uncover more information about the asset payoff than the market maker, and we interpret it as her skill as follows. At the beginning

³We abstract away from information acquisition by individual investors. Economies of scale would support this assumption as a natural equilibrium outcome, in which investors do not acquire the necessary skills and information, but rather remain uninformed and invest through a manager (e.g., ?). They do not directly participate in the market due to the lack of informational advantages over the market maker.

of the period, the manager draws a number s from a normal distribution, $N(0, \omega_s)$, where s is her private information. Leveraging s , she identifies an asset whose payoff is $\delta = \bar{\delta} + s$ with a publicly known mean payoff $\bar{\delta}$. Since the market maker does not observe s , the prior mean of δ from their perspective is $E[\delta] = \bar{\delta} + E[s] = \bar{\delta}$ and is biased by s from the manager's perspective. As the market maker would set a price $p = E[\delta] = \bar{\delta}$ in the absence of traders, s can be viewed as the asset's "potential mispricing."⁴ Indeed, as shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. Moreover, the variance $\text{Var}[s] = \omega_s$ controls the shape of the underlying distribution of s : a large ω_s represents a wider scope, making the manager more likely to identify an asset with a large mispricing, $|s|$. In light of this, ω_s is referred to as the manager's asset-selection *skill* with $|s|$ being its realized value.⁵ In practical terms, ω_s can reflect factors such as the manager's investment in higher education in finance and economics, which broadens the range of information available to her, while $|s|$ captures the realized ability to leverage such investments into superior decision-making in financial markets. In what follows, when there is no risk of confusion, we may refer to $|s|$ simply as "skill" without explicitly calling it "realized skill."

Investment. We model the fundraising and trading stages as a simultaneous game with rational expectations. On the one hand, the manager decides on the investment into the risky asset, anticipating the investors' and the market maker's reactions. This action is represented by a market order for x units of the asset sent to the market maker. In the financial market, noise traders also submit random order $u \sim N(0, \omega_u)$ exogenously. The market maker observes the aggregate order flow, $x + u$, and sets the semi-strong efficient price to earn

⁴ s is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

⁵This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., ?; ?).

zero expected profit:⁶

$$p = E[\delta|x + u]. \quad (2.1)$$

In raising funds from investors, the manager disseminates a noisy signal about her position to investors, $x + e$, where e represents a normally distributed noise, $e \sim N(0, \omega_e)$. This signal is delivered through private communication between the manager and her client investors and could capture the fund manager's practice of communicating a noisy signal about a high-conviction, concentrated risky position to investors to attract capital, consistent with the empirical evidence and theoretical rationale in ? and ?. It is informative about the manager's skill and the return of the fund, and investors use it in deciding on fund allocations. The noise variance, ω_e , is interpreted as the *opacity* of the fund. We first consider the baseline model with a perfectly transparent fund ($\omega_s = 0$), while Section XXX deals with general opacity ($\omega_s > 0$). Section XXX endogenizes ω_s through the manager's choice, e.g., she may obfuscates information by making trading strategies complex and incomprehensible to uninformed investors. Upon observing the fund information, $x + e$, investor i allocates m_i dollars to the fund, so that the fund's asset under management (AUM) becomes $M = \sum_{i=1}^n m_i$. The fund's fee structure is proportional to the total profits: the manager takes ϕ fraction of the total fund profit, while the remaining $1 - \phi$ fraction is distributed to individual investors according to their contributions to the fund, m_i/M .⁷

Maximization problem. With the trading profit from the risky investment being $\pi \equiv (\delta - p)x$, the total fund proceeding is expressed as

$$y = \pi + M. \quad (2.2)$$

⁶Assume that, on the equilibrium path, one competitive market maker trades the asset. As in ?, competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

⁷Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

The fund manager maximizes her fee income by controlling her investment strategy:

$$\max_x E[\phi y | s]. \quad (2.3)$$

Conversely, investor i maximizes her return from investment by controlling her fund contribution:

$$\max_{m_i} E \left[\frac{m_i}{M} (1 - \phi) y - m_i | x + e \right]. \quad (2.4)$$

Definition of equilibrium. Let $p(x + u)$ represent the market maker's pricing strategy, $x(s)$ denote the manager's trading strategy, and $\{m_i(x + e)\}_{i=1}^n$ denote investors' fund allocations. An equilibrium consists of $p(\cdot)$, $x(\cdot)$, and $\{m_i(\cdot)\}_{i=1}^n$ such that $p(\cdot)$ satisfies (??) given $x(\cdot)$, $x(\cdot)$ solves (??) given $p(\cdot)$ and $\{m_i(\cdot)\}_{i=1}^n$, and each $m_i(\cdot)$ solves (??) given $p(\cdot)$ and $x(\cdot)$.

3 Equilibrium

In this section, we search for a linear equilibrium under transparency ($\omega_s = 0$), in which each trader's market order x is linear in her skill, s . Proofs for the analytical results are provided in the Online Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \quad (3.1)$$

where $\beta > 0$ is an endogenous constant determined in the equilibrium. Conjecture (??) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.⁸ The scale factor β is referred to as the *trading intensity*.

⁸This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $E[\delta | s] - E[\delta] = s$.

3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow, $x + u$. Anticipating the trading strategy (??), the price in equation (??) can be rewritten as

$$p = \bar{\delta} + \lambda(x + u), \quad (3.2)$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \quad (3.3)$$

is “Kyle’s lambda,” a measure of the trader’s price impact.⁹ Following the literature, we define a market to be *liquid* if λ is small.

3.3 Delegation under Full Transparency

By applying $y = \pi + M$, investor i 's optimization problem in (??) is rewritten as

$$\max_{m_i} \frac{m_i}{M} (1 - \phi) E[\pi | x + e] - \phi m_i, \quad (3.4)$$

leading to the following optimal delegation:

$$m_i = \left(\frac{1 - \phi}{\phi} E[\pi | x + e] \sum_{j \neq i} m_j \right)^{\frac{1}{2}} - \sum_{j \neq i} m_j. \quad (3.5)$$

The cost of delegating investment is linear in the dollar value of the stake, as represented by the second term of (??), while the marginal benefit of an additional stake is diminishing, as is captured by the first term. Hence, given the total fund allocation by other investors, $\sum_{j \neq i} m_j$, investor i 's optimal delegation is determined by the FOC in equation (??).

We focus on the symmetric equilibrium where all investors provide the same

⁹From (??), we have $p = E[\delta | x + u] = E[E[\delta | x + u, s] | x + u] = E[\bar{\delta} + s | x + u]$. Since the market maker believes that the manager follows (??), s and $x + u$ are both normally distributed from the market maker's perspective. Thus, $p = \bar{\delta} + E[s] + \frac{\text{Cov}[s, x+u]}{\text{Var}[x+u]}(x + u - E[x + u]) = \bar{\delta} + \frac{\text{Cov}[s, \beta s + u]}{\text{Var}[\beta s + u]}(x + u - E[\beta s + u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u}(x + u)$.

amount of capital to the fund. By imposing $m_i = m_j$ for all $i, j \in \{1, 2, \dots, n\}$, (??) is reduced to

$$m = \frac{n-1}{n^2} \frac{1-\phi}{\phi} E[\pi|x+e]. \quad (3.6)$$

It suggests that the total asset under management (AUM) of the fund is

$$M \equiv nm = \frac{n-1}{n} \frac{1-\phi}{\phi} E[\pi|x+e]. \quad (3.7)$$

Individual and the total fund provisions are proportional to the expected value of trading profits conditional on the fund information disseminated to investors, $E[\pi|x+e]$. This is a natural consequence of the performance-based fee structure, as it makes the per-unit return of fund delegation proportional to the trading profit. In what follows, we denote the flow-performance sensitivity as $\theta \equiv \frac{n-1}{n} \frac{1-\phi}{\phi}$.

By incorporating the trading strategy in (??) and the pricing rule in (??), both the trading quantity, x , and the profit margin (i.e., informational advantages), $E[\delta - p|s]$, are proportional to the realized skill level, $|s|$, making the profit a quadratic function of s . When the fund information is perfectly transparent ($\omega_e = 0$), investors back out the realized skill level from the manager's trading strategy, $s = \frac{x}{\beta}$, so that the conditional expectation leads to the following estimate:

$$E[\pi|x+e] = (1-\lambda\beta)\beta E[s^2|s=x/\beta] = \frac{1-\lambda\beta}{\beta} x^2. \quad (3.8)$$

Together with the total fund provision in (??), the above estimate implies that the delegated investment may generate an additional incentive for the manager to control her trading strategy, x . Namely, the manager can positively influence the total fund flow by inflating the trading quantity, as it persuades investors of the manager's high realized skill level.

3.4 Manager's Optimization

By incorporating the total fund flow ([??] and [??]), the manager's problem in (??) is re-written as

$$\max_x \phi \left[sx - \lambda x^2 + \theta \frac{1 - \lambda\beta}{\beta} x^2 \right], \quad (3.9)$$

where the first two terms represents the fee income from the trading profit and are essentially identical to the original ? model. The third term arises due to the fund inflow and is determined by the quadratic size of risky investments. By taking the FOC, the optimal trading strategy involves

$$x = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)} s. \quad (3.10)$$

This strategy is consistent with our guess in (??) when the following condition holds:

$$\beta = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)}. \quad (3.11)$$

The solution is summarized by the following lemma:

Lemma 3.1. *When the fund information is transparent ($\omega_e = 0$), the equilibrium trading strategy and the pricing rules are specified by*

$$\beta = \sqrt{(1 + 2\theta) \frac{\omega_u}{\omega_s}}, \quad \lambda = \frac{1}{2(1 + \theta)} \sqrt{(1 + 2\theta) \frac{\omega_s}{\omega_u}}. \quad (3.12)$$

As is evident from problem (??), the flow-performance sensitivity, θ , captures the manager's additional incentive to control her risky investment arising from the fund delegation, and the equilibrium reduces to the original ? model as $\theta \rightarrow 0$. When $\theta = 0$, the manager faces a tradeoff in deciding on her trading intensity: while she seeks to exploit her informational advantage by trading more intensively, it reveals information to the market maker and raises the price impact, cutting into the trading profit. Without fund delegation ($\theta = 0$), the equilibrium β is determined to balance these competing effects. Conversely, $\theta > 0$ brings about the additional incentive: as shown in

(??) and (??), inflating the trading quantity benefits the manager by persuading investors of high $|s|$ and inducing a larger fund flow. When θ increases, the share of the total fund flow becomes larger in the manager’s utility, and this last channel becomes more significant. Therefore, β becomes an increasing function of θ . Conversely, the price impact, λ , decreases with θ , as the aggressive trading reveals too much private information to the market maker. From the market maker’s perspective, the information is “exaggerated” in that the order size is too large relative to the true $|s|$. Thus, she attempts to partially “undo” the manager’s information revelation by incorporating less of it into the price.¹⁰ This mechanism results in a reduced price impact.

This benchmark model and Lemma ?? isolate the fundamental mechanism through which the delegated investment structure inflates the manager’s investment behavior. The tendency of fund managers to “show off” their skill to attract capital is well discussed in the literature (e.g., ?; ?; ?), raising regulational concerns that fund managers have incentives to exaggerate the performance or quality of their funds.

Although the benchmark model of a transparent fund clearly generates the show-off incentives described above, it is important to ask whether this is a robust implication. In particular, as emphasized by ?, a skilled manager may also have incentives to conceal her skill, suggesting a fundamental tension between showing off and hiding. The more general model with opacity, developed in the next section, shows that, contrary to concerns raised by the literature and by policy makers, the manager may in fact prefer to hide her skill from *both* investors and the market.

4 Opaque Funds

This section introduces the fund opacity, $\omega_s > 0$, and demonstrates that belief-driven multiple equilibria show up regarding the trading strategy of the fund.

¹⁰This is because λ represents the price’s reaction to the order flow, $x + u$, rather than its reaction to the private information, s . The latter reaction is measured by $\beta\lambda$ and is monotonically increasing in β and θ .

We discuss economic implications drawn from the equilibrium analyses in Subsection ??.

4.1 Delegation under Fund Opacity

Assuming that $\omega_s > 0$, the analysis of an opaque fund largely follows the benchmark case in Section ?. In particular, investors' optimization and the structure of the resulting fund flow are identical to those in (??) and (??) in the benchmark model. However, due to the noisy revelation of the fund information, investors' conditional expectation about the fund return must be modified as follows:

Lemma 4.1. *Upon observing the fund information, $x + e$, investors' updated expected trading profit is given by*

$$E[\pi|x + e] = (1 - \lambda\beta)\beta(\hat{s}^2 + \hat{\omega}_s), \quad (4.1)$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = E[s|x + e] = \xi(x + e), \quad (4.2)$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_e}, \quad (4.3)$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \text{Var}[s|x + e] = \frac{\omega_s\omega_e}{\beta^2\omega_s + \omega_e}. \quad (4.4)$$

Investors update their belief about the realized skill using $x + e$, where ξ measures how much they rely on the fund information provided by the manager. Holding β constant, ξ is large when investors' prior is unreliable (ω_s is large) and the fund information is transparent (ω_e is small). In what follows, the updated expectation about the manager's realized skill, $\hat{s}$, is referred to as the *reputation*.

4.2 Manager's Optimization

Following the same procedures as the transparent benchmark, the manager's optimization problem in (??) is re-written as follows:

$$\max_x sx - \lambda x^2 + \theta(1 - \lambda\beta)\beta [\xi^2(x^2 + \omega_e) + \hat{\omega}_s]. \quad (4.5)$$

As in the baseline model, the first two terms of equation (??) represent the trader's expected trading profit. The third term corresponds to her profits from the fund flow, M , which also increases with $|x|$ because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell).

The unique solution to (??) (given β) is

$$x = F(\beta)s, \quad (4.6)$$

where

$$F(\beta) \equiv \frac{1}{2(\lambda - \theta(1 - \lambda\beta)\beta\xi^2)}. \quad (4.7)$$

$F(\cdot)$ is a nonlinear in β because λ and ξ are functions of β .

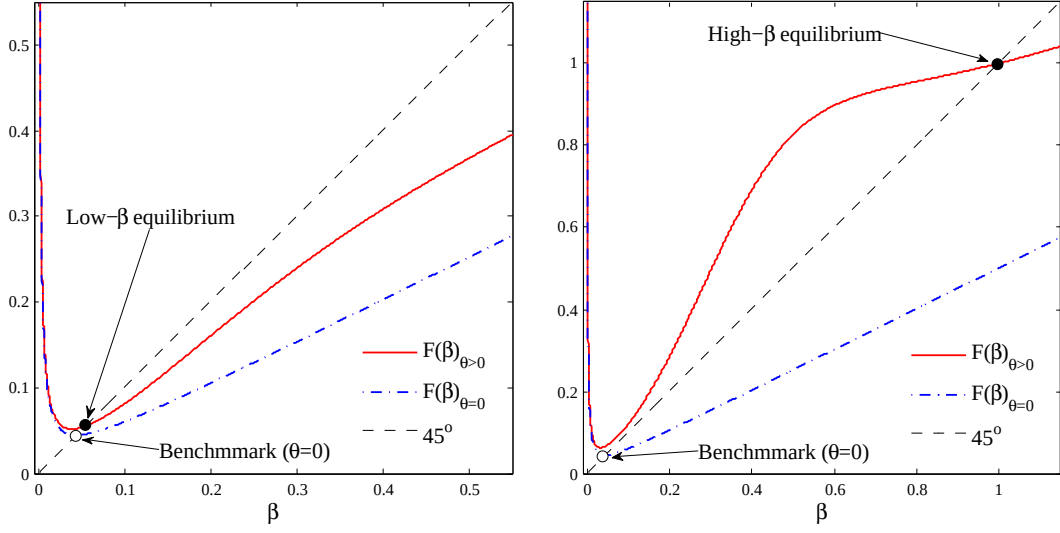
4.3 Equilibrium under Opacity

The manager optimally chooses (??) given that the market maker and investors believe that she follows strategy (??). In equilibrium, their beliefs must be correct, that is, (??) and (??) must be consistent with each other, requiring the following condition:

$$\beta = F(\beta). \quad (4.8)$$

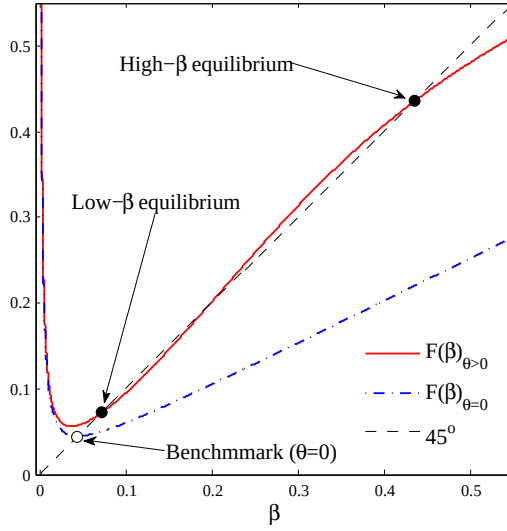
That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β , the equilibrium levels of x , p , and m are determined by equations (??), (??) and (??), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure ?? illustrates the determination of β for three different levels of fund opacity, ω_e . The intersections of $F(\beta)$ (the solid line) and the 45-degree



(a) Small θ ($\theta = 0.2$) $\Rightarrow$ unique low- β equilibrium.

(b) Large θ ($\theta = 0.55$) $\Rightarrow$ unique high- β equilibrium.



(c) Intermediate θ ($\theta = 0.33$) $\Rightarrow$ multiple high- β & low- β equilibria.

Figure 1: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ . The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, and $\omega_u = 0.1$.

line yield the equilibrium levels of β . We also plot $F(\beta)$ with $\theta = 0$ (the dash-dotted line) and denote its solution as $\beta_{\theta=0}$ to highlight the original model of ? where fund delegation is irrelevant.

Panel (a) of Figure ?? shows that if ω_e is large, there is a unique “low- β equilibrium” where β is close to $\beta_{\theta=0}$. Panel (b) shows that a large ω_e leads to a unique “high- β equilibrium,” in which β is much larger than $\beta_{\theta=0}$. Panel (c) shows that multiple equilibria arise for an intermediate level of ω_e . Specifically, the high- β and the low- β equilibria are stable, as $F(\beta)$ crosses the 45-degree line from above, whereas the one with an intermediate β is unstable, as $F(\beta)$ crosses the line from below.¹¹ In what follows, we focus only on the stable equilibria.

Impact of fund opacity. To see the intuition behind the different equilibrium patterns presented in Figure ??, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in ?, this motive leads the trader to decrease β . On the other hand, she wishes to trade intensively to signal her high skill to her client investors, aiming to establish a strong reputation and attract a large fund flow. This motive encourages her to increase β . In summary, she faces a tradeoff between “hiding” her skill from the market maker and “showing it off” to her clients. The fund opacity, ω_e , governs this tradeoff by influencing the sensitivity of the manager’s reputation, $\hat{s}$, to her investment behavior, x . In particular, Lemma ?? demonstrates that the fund flow becomes insensitive to x under opacity, as it becomes difficult for investors to estimate the skill based on the fund information, weakening the showing-off motive.

For a opaque fund with a large ω_e (panel (a)), the hiding motive dominates the showing-off motive, leading to a unique low- β equilibrium. In this equilib-

¹¹We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents’ belief about β , meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents’ beliefs.

rium, client investors believe that the manager’s trading intensity is relatively low. Knowing this, the manager could inflate her reputation by placing an order larger than what her clients anticipates. However, she would refrain from doing so because it would cause a large price impact, reducing the fund’s investment profits. Consequently, the equilibrium β is close to $\beta_{\theta=0}$, meaning that it is a “Kyle-like equilibrium.”

For a transparent fund with a small ω_e (panel (b)), the showing-off motive prevails, resulting in a unique high- β equilibrium. In this equilibrium, the manager is willing to move the price substantially, undermining investment profits to appear skilled. Importantly, client investors’ learning is not manipulated, as their belief about the manager’s trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her clients to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her clients’ belief. To emphasize this mechanism, we call this equilibrium the “reputation-driven equilibrium.”

For intermediate levels of ω_e (panel (c)), there is no clear dominance between the two motives, hence both the low- β and the high- β equilibria are supported under intermediate fund opacity. Each equilibrium is associated with a self-fulfilling belief of the market maker and investors.

The following proposition and Figure ?? summarize the above discussion and provide an analytical characterization of the equilibria.

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b1 endstream endobj

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